INCH-POUND

MIL-DTL-55302/176C
14 December 2007

SUPERSEDING
MIL- DTL-55302/176B
13 September 2002

DETAIL SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: 3 ROW, PIN ASSEMBLIES, 75 THROUGH 405 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.

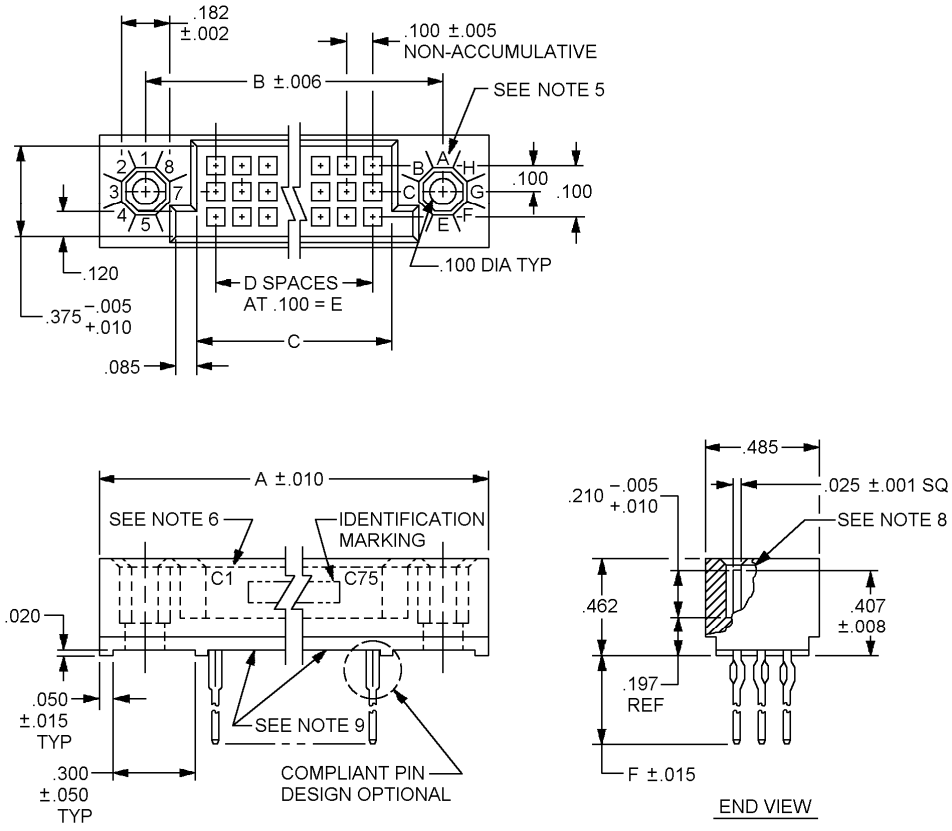
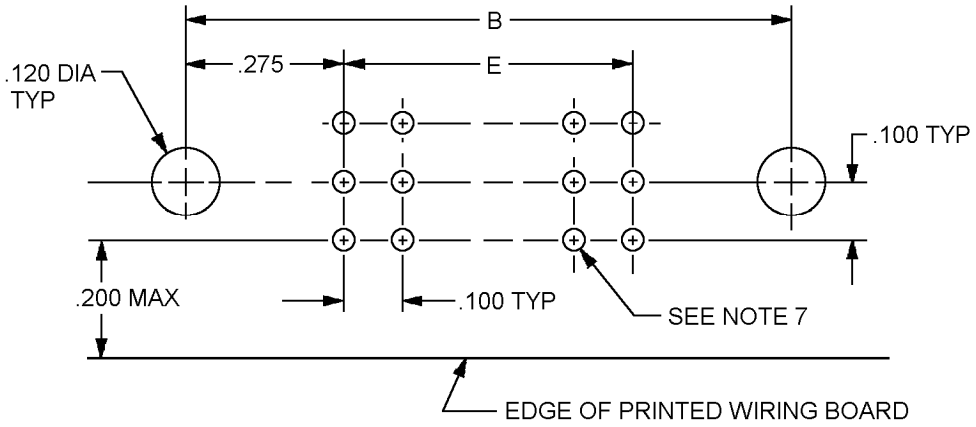


FIGURE 1. Connectors, pin assemblies .100 inch (2.54 mm) spacing, 75 through 225 contacts.

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Recommended circuit layout

F ± .015 (0.38 mm)		Type	Type of termination
Inches	mm		
.733	18.62	I	Compliant pin (wrappost)
.533	13.54	II	
.250	6.35	IV	
.190	4.83	VI	Solder post (see note 10)

Inches	mm	Inches	mm	Inches	mm
.001	0.03	.025	0.64	.275	6.98
,002	0.05	.050	1.27	.300	7.62
.005	0.13	.085	2.16	.375	9.52
.006	0.15	.100	2.54	.407	10.34
.008	0.20	.120	3.05	.462	11.73
.010	0.25	.182	4.62	.485	12.32
.015	0.38	.197	5.00		
.020	0.51	.210	5.33		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ± .005 inch (0.13 mm) on decimals and ± 2° on angles.
4. These connectors mate with connectors in accordance with MIL-DTL-55302/177.
5. Location indicators embossed on surface.
6. Numbers and indicating end cavities, letters indicating row nearest to polarizing feature, and markings every ten positions stamped on this surface.
7. Holes for solder type shall be .042 - .050 inch (1.07 – 1.27 mm). Holes for compliant pin shall be manufactured as follows: .0453 ± .0010 inch (1.151 ±0.025 mm) drilled hole (1.15 mm drill) .001 - .003 inch (0.03 – 0.08 mm) thick copper on wall, .0003 inch (0.008 mm) minimum tin lead plating, finished hole diameter .037 - .043 (0.94 – 1.09 mm) after plating, .036 - .043 inch (0.91 – 1.09 mm) after reflow, copper hardness (knoop) 150 maximum.
8. Chamfers shall be .015 inch (0.38 mm) minimum at a 45° angle ± 5°.
9. There shall be a minimum of two standoffs per inch of connector length for the balance of the part. Standoffs shall not compromise the terminating area washout feature.
10. Solder post shall be a .036 (0.91 mm) maximum diameter or .036 inch (0.91 mm) maximum across the diagonal.

FIGURE 1. Connectors, pin assemblies .100 inch (2.54 mm) spacing, 75 through 225 contacts – Continued.

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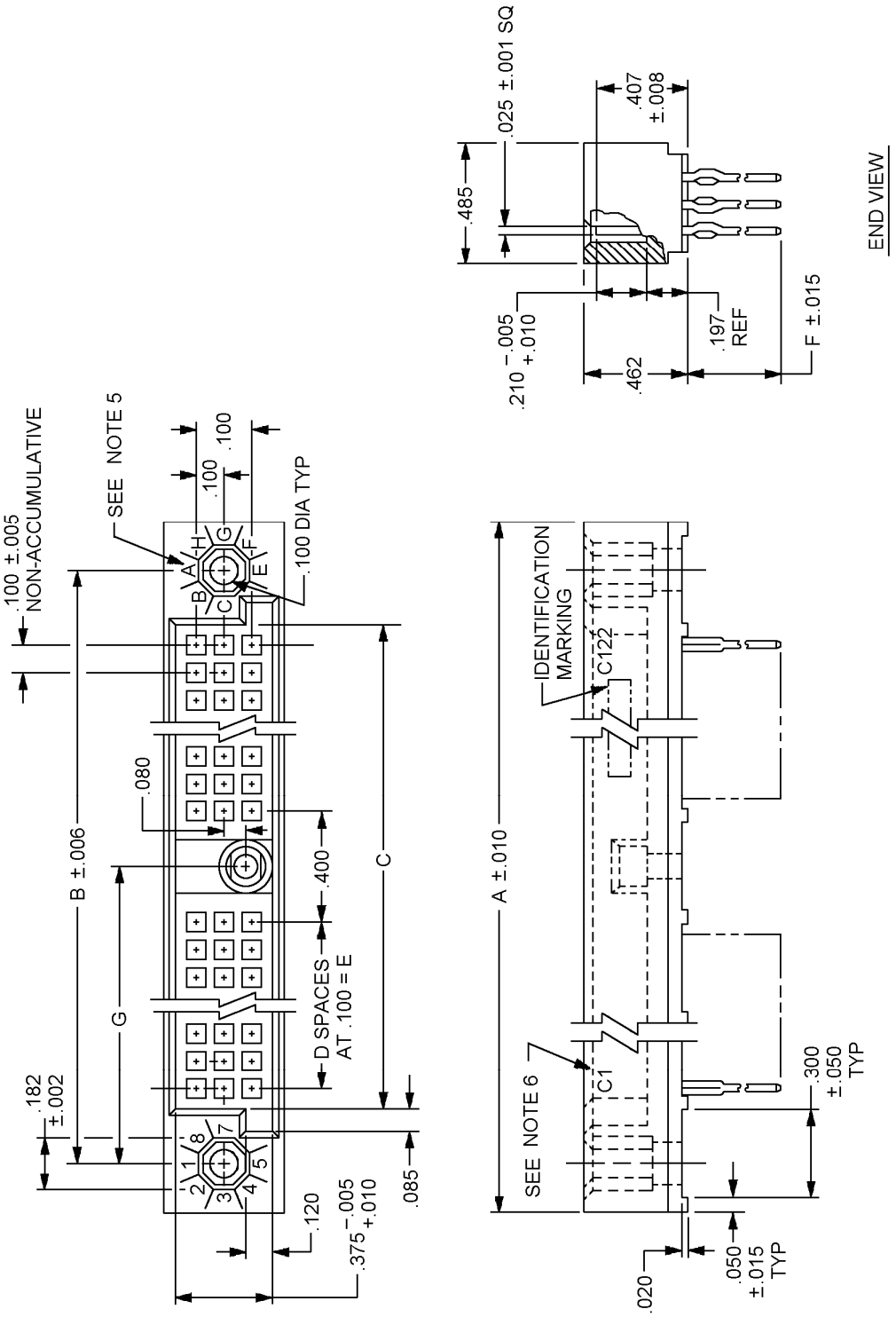
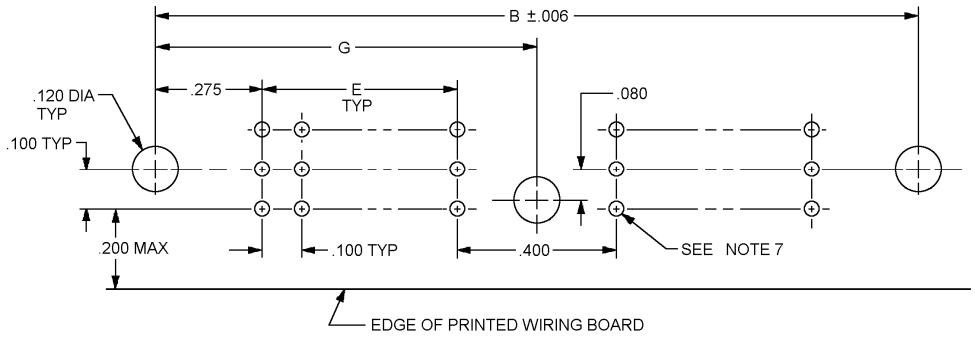


FIGURE 2. Connectors, pin assemblies .100 (2.54 mm) spacing with one guide, 240 through 366 contacts.

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Recommended circuit layout

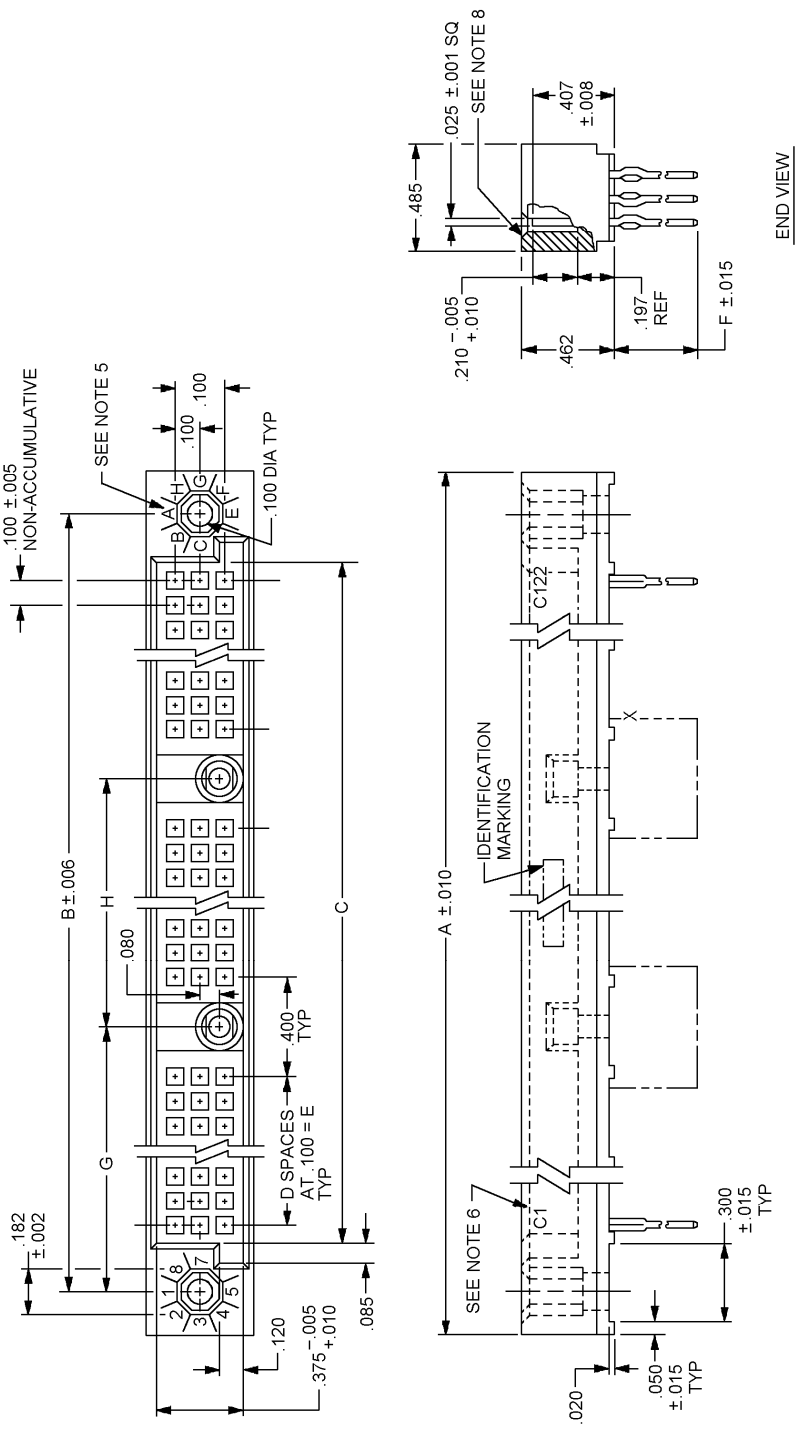
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.008	0.20	.100	2.54	.407	10.34
.010	0.25	.120	3.05	.462	11.73
.015	0.38	.182	4.62	.485	12.32
.016	0.41	.197	5.00		
.020	0.51	.210	5.33		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ± .005 inch (0.13 mm) on decimals and ± 2° on angles.
4. Guide pins are purchased separately. See MIL-DTL-55302/182.
5. Location indicators embossed on surface.
6. Numbers and indicating end cavities, letters indicating row nearest to polarizing feature, and markings every ten positions stamped on this surface.
7. Holes for solder type shall be .042 - .050 inch (1.07 – 1.27 mm). Holes for compliant pin shall be manufactured as follows: .0453 ± .0010 inch (1.151 ±0.025 mm) drilled hole (1.15 mm drill) .001 - .003 inch (0.03 – 0.08 mm) thick copper on wall, .0003 inch (0.008 mm) minimum tin lead plating, finished hole diameter .037 - .043 (0.94 – 1.09 mm) after plating, .036 - .043 inch (0.91 – 1.09 mm) after reflow, copper hardness (knoop) 150 maximum.
8. Chamfers shall be .015 inch (0.38 mm) minimum at a 45° angle ± 5°.
9. There shall be a minimum of two standoffs per inch of connector length for the balance of the part. Standoffs shall not compromise the terminating area washout feature.
10. Solder post shall be a .036 (0.91 mm) maximum diameter or .036 (0.91 mm) maximum across the diagonal.

FIGURE 2. Connectors, pin assemblies .100 inch (2.54 mm) spacing with one guide, 240 through 366 contacts - Continued.

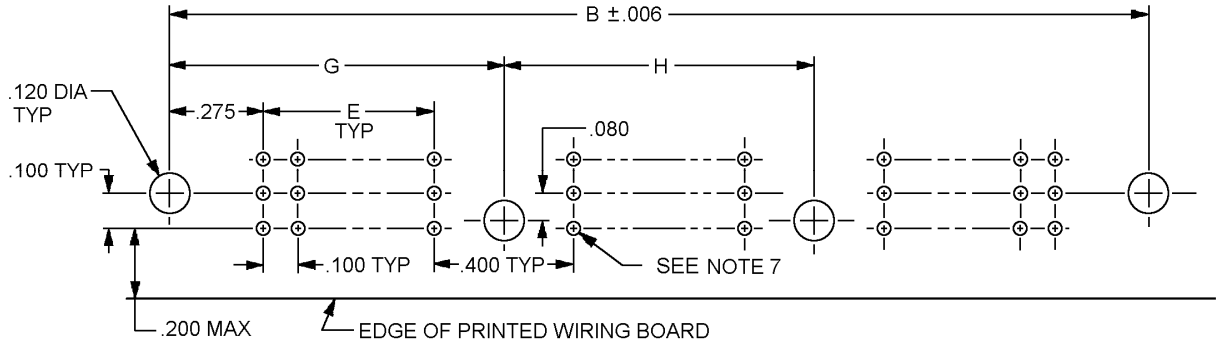
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Recommended circuit layout

FIGURE 3. Connectors, pin assemblies .100 (2.54 mm) spacing with two guides, 387 through 405 contacts.

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F ± .015 (0.38 mm)		Type	Type of termination
Inches	mm		
.733	18.62	I	Compliant pin (wrappost)
.533	13.54	II	
.250	6.35	IV	
.190	4.83	VI	Solder post (see note 10)

Inches	mm	Inches	mm	Inches	mm
.001	0.03	.025	0.64	.210	5.33
.002	0.05	.050	1.27	.275	6.98
.005	0.13	.080	2.03	.300	7.62
.006	0.15	.085	2.16	.375	9.52
.008	0.20	.100	2.54	.400	10.16
.010	0.25	.120	3.05	.407	10.34
.015	0.38	.182	4.62	.462	11.73
.020	0.51	.197	5.00	.485	12.32

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm) on decimals and ± 2° on angles.
4. Guide pins are purchased separately. See MIL-DTL-55302/182.
5. Location indicators embossed on surface.
6. Numbers and indicating end cavities, letters indicating row nearest to polarizing feature, and markings every ten positions stamped on this surface.
7. Holes for solder type shall be .042 - .050 inch (1.07 – 1.27 mm). Holes for compliant pin shall be manufactured as follows: .0453 ± .0010 inch (1.151 ±0.025 mm) drilled hole (1.15 mm drill) .001 - .003 inch (0.03 – 0.08 mm) thick copper on wall, .0003 inch (0.008 mm) minimum tin lead plating, finished hole diameter .037 - .043 (0.94 – 1.09 mm) after plating, .036 - .043 inch (0.91 – 1.09 mm) after reflow, copper hardness (knoop) 150 maximum.
8. Chamfers shall be .015 inch (0.38 mm) minimum at a 45° angle ± 5°.
9. There shall be a minimum of two standoffs per inch of connector length for the balance of the part. Standoffs shall not compromise the terminating area washout feature.
10. See figure 1 for solder post type. Solder post shall be a .036 (0.91 mm) maximum diameter or .036 (0.91 mm) maximum across the diagonal.

FIGURE 3. Connectors, pin assemblies .100 inch (2.54 mm) spacing with one guide, 387 through 405 contacts - Continued.

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TABLE I. Dash number, number of contacts, contact type, and dimensions. 1/ 2/ 3/

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
01 02 04 06	75	1	I II IV VI	3.250 (82.55)	2.950 (74.93)	2.550 (64.77)	24	2.400 (60.96)			1-25
19 20 22 24	96	1	I II IV VI	3.950 (100.33)	3.650 (92.71)	3.250 (82.55)	31	3.100 (78.74)			1-32
37 38 40 42	120	1	I II IV VI	4.750 (120.65)	4.450 (113.03)	4.050 (102.87)	39	3.900 (99.06)			1-40
55 56 58 60	150	1	I II IV VI	5.750 (146.05)	5.450 (138.43)	5.050 (128.27)	49	4.900 (124.46)			1-50
64 65 67 69	165	1	I II IV VI	6.250 (158.75)	5.950 (151.13)	5.550 (140.97)	54	5.400 (137.16)			1-55
73 74 76 78	180	1	I II IV VI	6.750 (171.45)	6.450 (163.83)	6.050 (153.67)	59	5.900 (149.86)			1-60
91 92 94 96	201	1	I II IV VI	7.450 (189.23)	7.150 (181.61)	6.750 (171.45)	66	6.600 (167.64)			1-67
100 101 103 105	210	1	I II IV VI	7.750 (196.85)	7.450 (189.23)	7.050 (179.07)	69	6.900 (175.26)			1-70
118 119 121 123	240	2	I II IV VI	9.050 (229.87)	8.750 (222.25)	8.350 (212.09)	39	3.900 (99.06)	4.375 (111.13)		1-80

See footnote at end of table.

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TABLE I. Dash number, number of contacts, contact type, and dimensions – Continued. 1/ 2/

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
136 137 139 141	276	2	I II IV VI	10.250 (260.35)	9.950 (252.73)	9.550 (242.57)	45	4.500 (114.30)	4.975 (126.37)		1-92
163 164 166 168	330	2	I II IV VI	12.050 (306.07)	11.750 (298.45)	11.350 (288.29)	54	5.400 (137.16)	5.875 (149.23)		1-110
181 182 184 186	366	2	I II IV VI	13.250 (336.55)	12.950 (328.93)	12.550 (318.77)	60	6.000 (152.40)	6.475 (164.47)		1-122
199 200 202 204	405	2	I II IV VI	14.850 (377.19)	14.550 (369.57)	14.150 (359.41)	44	4.400 (111.76)	4.875 (123.83)	4.800 (121.92)	1-135

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.

TABLE II. Contact plating requirements. 1/

Type	Finish requirement
A	Overall contact finish: Shall be gold in accordance with MIL-DTL-55302.
B	Localized contact finish. Contact engagement area, .170-inch minimum length, shall be gold over nickel in accordance with MIL-DTL-55302. Compliant pin area (contact type I, .748 inch minimum length, type II, .548 minimum length; type IV, .265 inch minimum length) shall be tin-lead over nickel in accordance with MIL-DTL-55302. Solder post area contact type VI, .205 minimum length, shall be tin-lead over nickel in accordance with MIL-DTL-55302.

1/ Type C finish is deleted and superseded by type A or B.

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REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figures 1, 2, 3 and table I.

Materials: In accordance with MIL-DTL-55302.

Plating: See table II.

Current rating: 3.0 amperes maximum per contact, 2.25 amperes continuous per contact at room ambient with no more than two adjacent contacts carrying this current.

Keying (see MIL-DTL-55302/31):

Two keys, Part or Identifying Number (PIN) M55302/31-04, and two .086-56 UNC-2A mounting screws are recommended. Keys and screws shall be ordered separately.

Jackscrews (see MIL-DTL-55302/182):

Use M55302/182-11, -12, -13, -14, -15, or -16. Jackscrews are purchased separately.

Mating and unmating: The mating force in pounds shall be the number of contacts multiplied by .250; the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: No individual contact pair shall have a resistance exceeding .015-ohm initial and .020 after testing.

Contact retention: 3 pounds, minimum.

Dielectric withstanding voltage:

Sea level: 900 volts rms.

70,000 feet: 200 volts rms.

2 milliamperes maximum leakage current.

PIN: M55302/176 (and dash number from table I and type from table II).

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Group qualification: See table III.

TABLE III. Group qualification.

Qualification of any of the following connectors 1/ 2/	Qualifies the following connectors	Retains qualification on the following
M55302/173-*** M55302/175-***	M55302/173-*** M55302/175-***	M55302/173-*** M55302/175-*** M55302/176-*** M55302/178-*** M55302/179-*** M55302/181-***
M55302/176-*** M55302/178-***	M55302/173-*** M55302/175-*** M55302/176-*** M55302/178-***	
M55302/179-*** M55302/181-***	M55302/173-*** M55302/175-*** M55302/176-*** M55302/178-*** M55302/179-*** M55302/181-***	

1/ For initial qualification, connectors with the largest number of contact positions from the left hand column shall be used to obtain qualification for parts in corresponding center column.
2/ For qualification retention, data may be supplied on any two parts from the left hand column in order to retain qualification for the parts listed in the right hand column.

CHANGES FROM PREVIOUS ISSUE: The margins of this specification are marked with vertical lines to indicate where changes from previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-55302/31
MIL-DTL-55302/177
MIL-DTL-55302/182

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2007-034)

Review activities:
Army - AR, AT, AV, CR4, MI
Navy - AS, MC, SH
Air Force - 19, 99

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